DELEGATION THROUGH ATTACHED POLICY SETS

The delegation model presented here considers delegated authority as structured data stored in a delegation record. Policies evaluate this data at runtime to determine if a subagent is allowed to act within the delegated scope. In this way, delegation is expressed through *policy as data*: the authority being transferred is represented by attributes that policies interpret.

Other delegation models represent delegated authority directly through *policy as code*.

In these systems, delegation happens when a trusted component creates or attaches policies that specify what a subagent is allowed to do. Instead of recording that one agent has delegated particular permissions to another, the system grants authority by linking a set of policies to the subagent or to a role it assumes. This method is similar to that used by cloud platforms like Amazon Web Services, in which permissions are granted by attaching identity-based or resource-based policies to roles and service principals.

This model generally assumes a default posture in which all actions are forbidden unless explicitly allowed. Delegated authority is added by attaching narrowly scoped policy statements that allow specific operations. These policies can be temporary, contextual, or dynamically created during task execution.

Both methods make authority transparent and enable easier delegation. Delegation as data allows flexible, task-specific authorization because policies assess contextual attributes during execution. Delegation with attached policy sets helps clarify permissions by making authority clear through explicit policy artifacts.

In practice, systems might combine these approaches by using policy attachment for coarse-grained delegation and structured delegation data for dynamic, fine-grained control.